Department of Computer Science and Engineering
Academic Year 2024 – 2025 (Even Semester)

Degree, Semester & Branch: IV Semester B.E. CSE 'B'
Course Code & Title: CS3491 Artificial Intelligence and Machine Learning
Name of the Faculty member (s): Dr.B.Vijayalakshmi, AP (SG)/CSE

Innovative Practice Description

- **Unit / Topic: Unit IV / Unsupervised & Ensemble Learning**
- **Course Outcome:** CO4
- **Topic Learning Outcome: 4a,c**
- **Activity Chosen: Concept Map**
- **Justification:**
  The basic understanding and classification of ensemble learning and unsupervised machine learning models are very much essential for solving real world applications. Hence a concept map activity is chosen to provide a way to summarize the different machine learning techniques.
- **Time Allotted for the Activity: 10 Minutes**
- **Details of the Implementation:**
  - The activity was conducted at the last 10 minutes of the class after discussing the various object oriented programming concepts.
  - The students are asked to prepare a concept map by depicting the different concepts involved in object oriented programming.
  - The students recollect the concepts involved in OOPS and present the concepts in a easily understandable way.
  - Fig.1 shows the concept maps drawn by the students.
  - The activity was conducted during the final 10 minutes of the class to conclude the discussion of various machine learning techniques, including supervised, unsupervised, and ensemble learning.
  - The students are instructed to create a concept map that illustrates the various machine learning techniques.
  - The students recall the various machine learning methods and articulate the concepts in a manner that is readily comprehensible.
  - The students' concept maps are illustrated in Fig. 1.

- **CO – PO / PSO mapping:**

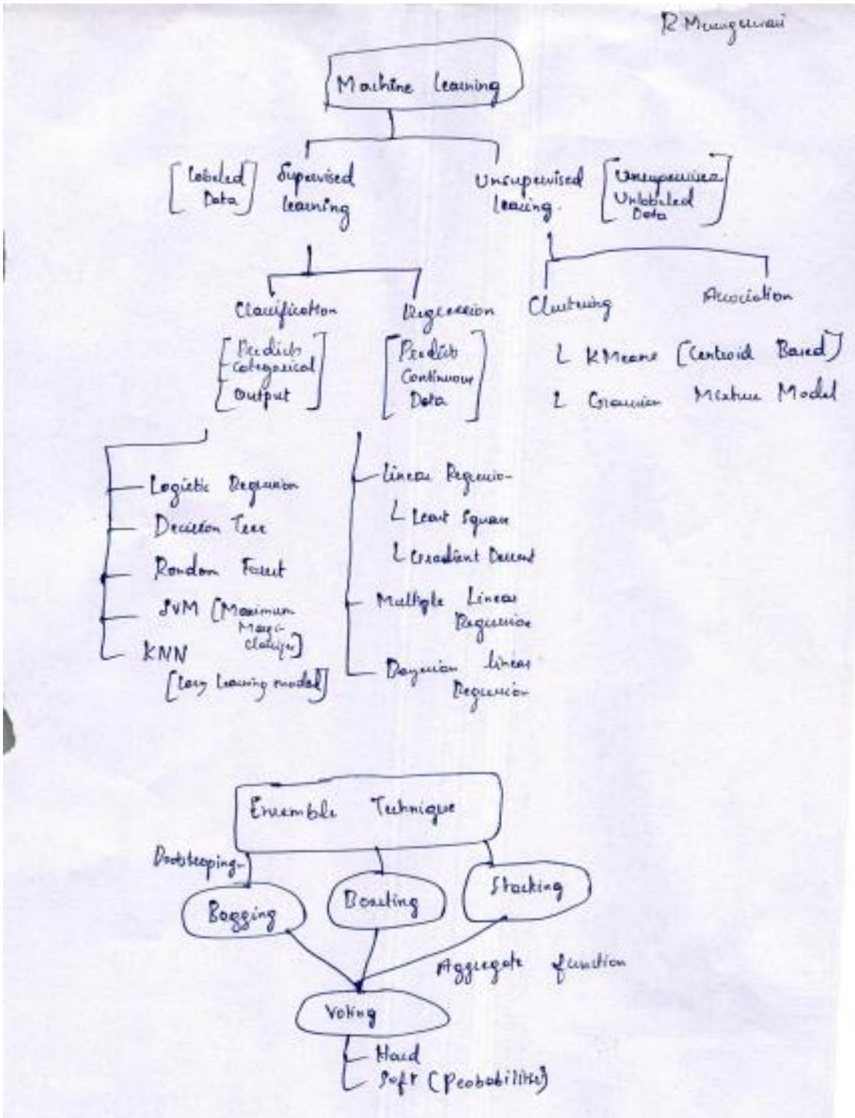
CO	PO1	PO2	PO3	PO4	PO5	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO4	3	3	2	2	3	1	2	2	1	2	1	1	3

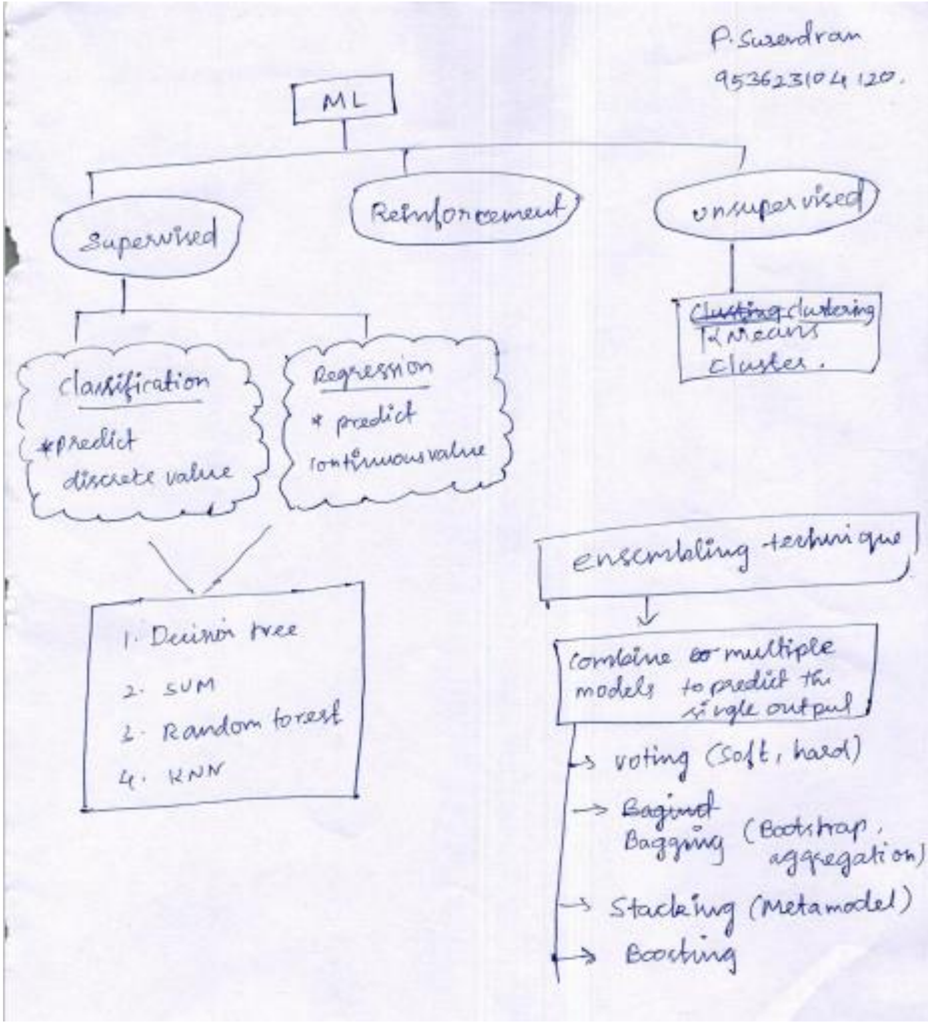
(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO9	PO10	PSO3
	3	**2**	**1**	**1**	**3**
Justification for correlation	The students have the knowledge of machine learning algorithms	The students categorize different ML techniques.	The students individually present their ideas.	The students communicate the Ml techniques in the form of concept map	The ML algorithms are used in solving real world problems in industry.

- **Images / Screenshot of the practice:**




P. Surendran
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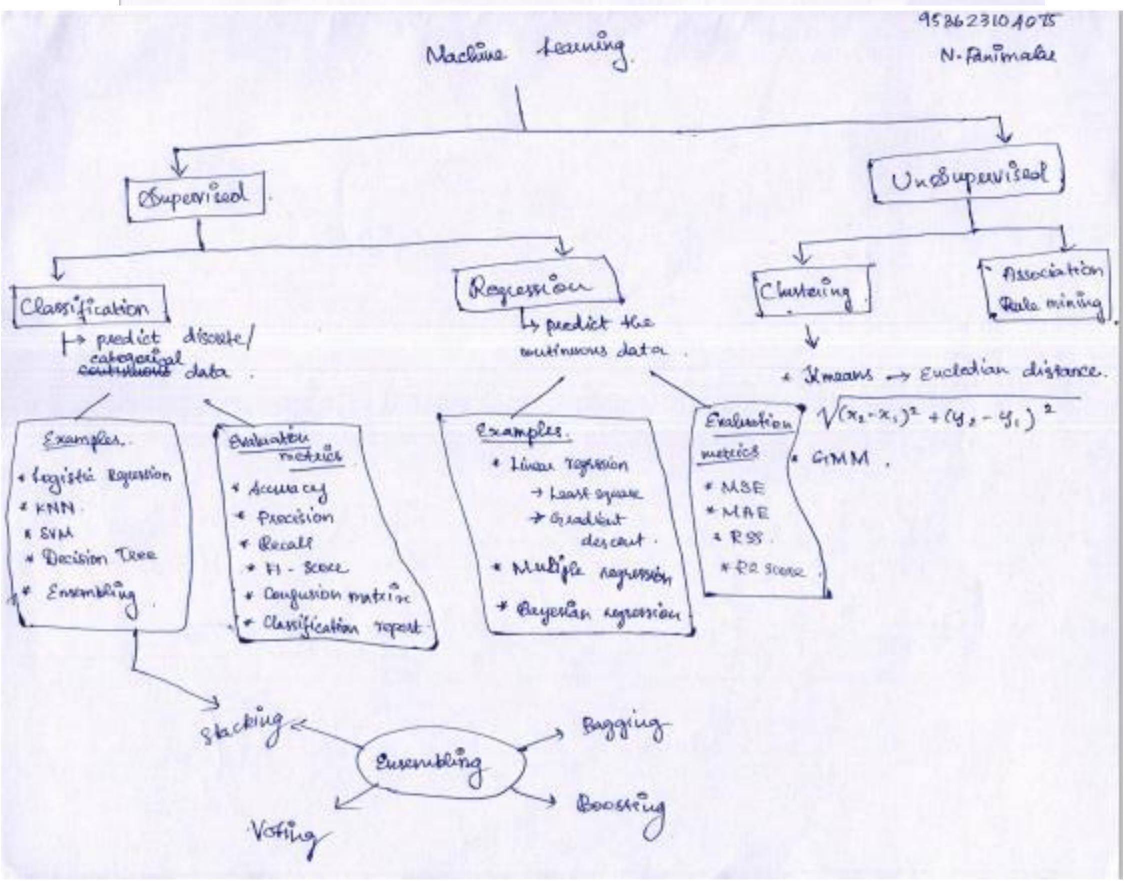
ML
Supervised
Reinforcement
unsupervised
~~Clustring~~ clustering
K Means
Cluster.
classification
* predict
discrete value
Regression
* predict
continuous value
1. Decision tree
2. SVM
3. Random forest
4. KNN
ensembling technique
combine ~~or~~ multiple
models to predict the
single output
voting (soft, hard)
~~Bagined~~
Bagging (Bootstrap, aggregation)
Stacking (Metamodel)
Boosting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N. Panimalar
Machine Learning.
Supervised
UnSupervised
Classification
predict discrete/ categorical ~~continuous~~ data.
Regression
predict the continuous data.
Clustering
Association Rule mining
Kmeans → Euclidean distance.
$\sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$
GMM.
Examples.
* Logistic Regression
* KNN.
* SVM
* Decision Tree
* Ensembling.
Evaluation metrics
* Accuracy
* Precision
* Recall
* F1 - Score
* Confusion matrix
* Classification report
Examples.
* Linear regression
→ Least square
→ Gradient descent.
* Multiple regression
* Bayesian regression
Evaluation metrics
* MSE
* MAE
* RSS.
* R2 score.
stacking
Bagging
Ensembling
Voting
Boosting

Fig. 1: Sample Concept Map

- **Reflective Critique:**
  - ***Feedback of practice from students and other stakeholders:***
    - The students are all highly enthusiastic about creating a concept map that depicts the different ML techniques.
  - ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)
    - This activity enables them to organize the entire concept in a single visual.
    - Their comprehension can be assessed by examining the concept map they have created.
  - ***Challenges faced in implementation:***
    - The completion of the map may require additional time for certain students.
    - Some student's representations on the map are limited to a few topics.

References:

- https://www.schrockguide.net/concept-mapping.html